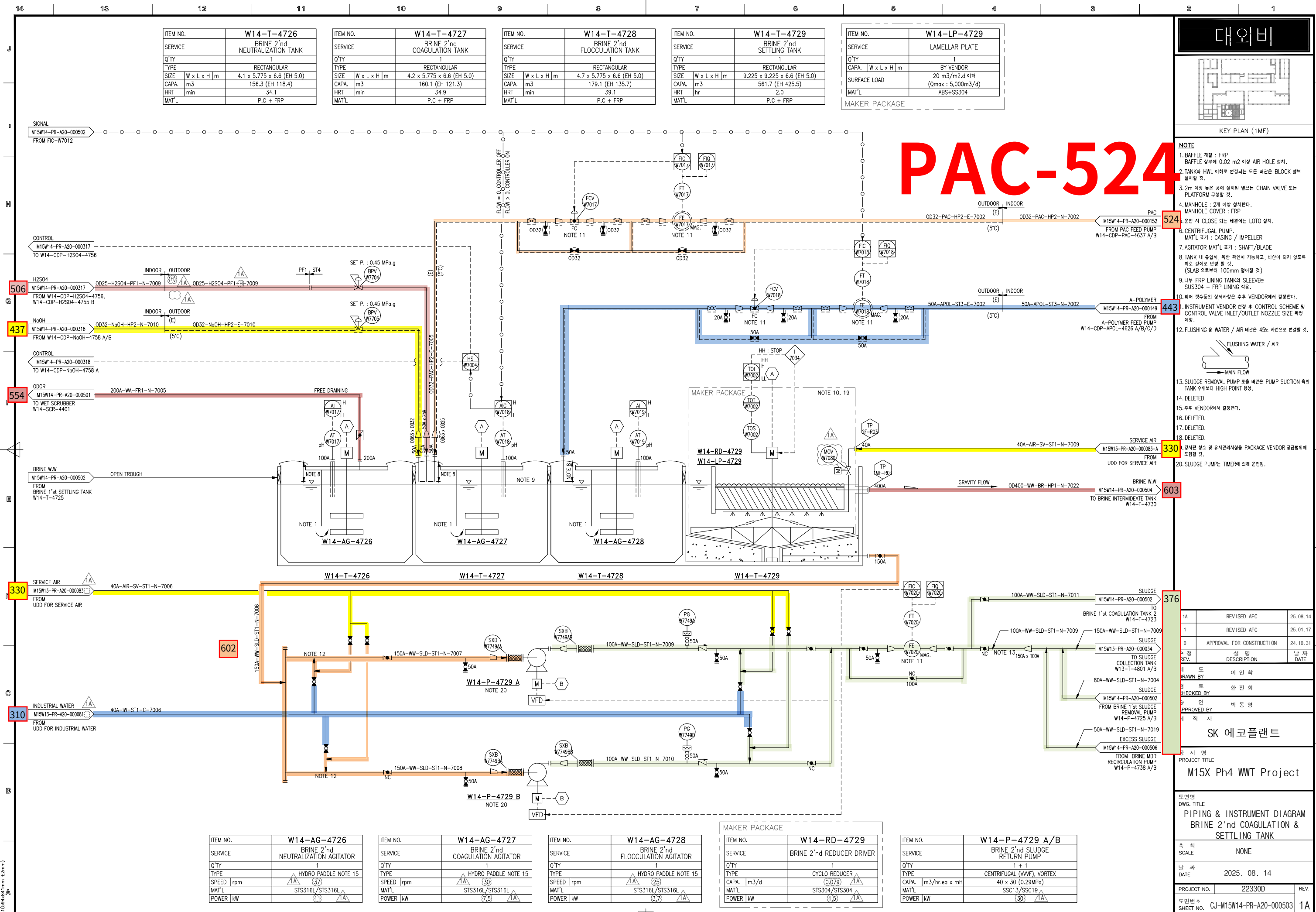


# PAC-524

**PROJECT TITLE : M15X Ph2 Project**

| No | PKG No | UNIT | LINE NUMBER            |        |     | P&ID                    |     | REMARK |
|----|--------|------|------------------------|--------|-----|-------------------------|-----|--------|
|    |        |      | ISO DRAWING No         | SHEET  | REV | P&ID No                 | REV |        |
| 1  | 524    | R2FA | OD32-PAC-HP2-E-7002-01 | 1 OF 1 | 1   | CJ-M15W14-PR-A20-000152 |     |        |
| 2  | 524    | R2FA | OD32-PAC-HP2-E-7005-01 | 1 OF 1 | 1   | CJ-M15W14-PR-A20-000503 |     |        |
| 3  | 524    | 2FA  | OD32-PAC-HP2-E-7002-01 | 1 OF 1 | 1   | CJ-M15W14-PR-A20-000152 |     |        |
| 4  |        |      |                        |        |     |                         |     |        |
| 5  |        |      |                        |        |     |                         |     |        |
| 6  |        |      |                        |        |     |                         |     |        |
| 7  |        |      |                        |        |     |                         |     |        |
| 8  |        |      |                        |        |     |                         |     |        |
| 9  |        |      |                        |        |     |                         |     |        |
| 10 |        |      |                        |        |     |                         |     |        |
| 11 |        |      |                        |        |     |                         |     |        |
| 12 |        |      |                        |        |     |                         |     |        |



대외비

KEY PLAN (1MF)

- NOTE
1. BAFFLE 제형 : FRP  
BAFFLE 상부에 0.02 m 이상 AIR HOLE 설치.
  2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
  3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
  4. MANHOLE : 2개 이상 설치한다.  
MANHOLE COVER : FRP
  5. 온천 시 CLOSE 되는 배관은 LOTO 설치.
  6. CENTRIFUGAL PUMP.  
MAT'L 표기 : CASING / IMPELLER
  7. AGITATOR MAT'L 표기 : SHAFT/BLADE
  8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.  
(SLAB으로부터 100mm 떨어질 것)
  9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
  10. 뒤어 것수등의 상세사상은 추후 VENDOR에서 결정한다.
  11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
  12. FLUSHING 용 WATER / AIR 배관은 45도 서언으로 연결할 것.
  13. SLUDGE REMOVAL PUMP 또는 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 향상.
  14. DELETED.
  15. 추후 VENDOR에서 결정한다.
  16. DELETED.
  17. DELETED.
  18. DELETED.
  19. 경사면 장소 및 유지관리시설을 PACKAGE VENDOR 공급범위에 포함할 것.
  20. SLUDGE PUMP는 TIMER에 의해 운전됨.

|             |                           |          |
|-------------|---------------------------|----------|
| 1A          | REVISED AFC               | 25.08.14 |
| 1           | REVISED AFC               | 25.01.17 |
| 0           | APPROVAL FOR CONSTRUCTION | 24.10.31 |
| 수정          | 실명                        | 날짜       |
| REV.        | DESCRIPTION               | DATE     |
| 도           | 이인학                       |          |
| 검           | 토한진희                      |          |
| HECKED BY   |                           |          |
| 승인          | 박동명                       |          |
| APPROVED BY |                           |          |
| 작사          |                           |          |

SK 에코플랜트

공 사 명  
PROJECT TITLE

M15X Ph4 WWT Project

도면명  
DWG. TITLE

PIPING & INSTRUMENT DIAGRAM  
BRINE 2<sup>nd</sup> COAGULATION &  
SETTLING TANK

속 척  
SCALE

NONE

날 짜  
DATE

2025. 08. 14

PROJECT NO.

22330D

REV.

A

도면번호  
SHEET NO.

CJ-M15W14-PR-A20-000503

1A

|          |                                              |
|----------|----------------------------------------------|
| ITEM NO. | W14-T-4726                                   |
| SERVICE  | BRINE 2 <sup>nd</sup><br>NEUTRALIZATION TANK |
| Q'TY     | 1                                            |
| TYPE     | RECTANGULAR                                  |
| SIZE     | W x L x H   m<br>4.1 x 5.775 x 6.6 (EH 5.0)  |
| CAPA.    | m <sup>3</sup><br>156.3 (EH 118.4)           |
| HRT      | min<br>34.1                                  |
| MAT'L    | P.C + FRP                                    |

|          |                                             |
|----------|---------------------------------------------|
| ITEM NO. | W14-T-4727                                  |
| SERVICE  | BRINE 2 <sup>nd</sup><br>COAGULATION TANK   |
| Q'TY     | 1                                           |
| TYPE     | RECTANGULAR                                 |
| SIZE     | W x L x H   m<br>4.2 x 5.775 x 6.6 (EH 5.0) |
| CAPA.    | m <sup>3</sup><br>160.1 (EH 121.3)          |
| HRT      | min<br>34.9                                 |
| MAT'L    | P.C + FRP                                   |

|          |                                             |
|----------|---------------------------------------------|
| ITEM NO. | W14-T-4728                                  |
| SERVICE  | BRINE 2 <sup>nd</sup><br>FLOCCULATION TANK  |
| Q'TY     | 1                                           |
| TYPE     | RECTANGULAR                                 |
| SIZE     | W x L x H   m<br>4.7 x 5.775 x 6.6 (EH 5.0) |
| CAPA.    | m <sup>3</sup><br>179.1 (EH 135.7)          |
| HRT      | min<br>39.1                                 |
| MAT'L    | P.C + FRP                                   |

|          |                                               |
|----------|-----------------------------------------------|
| ITEM NO. | W14-T-4729                                    |
| SERVICE  | BRINE 2 <sup>nd</sup><br>SETTLING TANK        |
| Q'TY     | 1                                             |
| TYPE     | RECTANGULAR                                   |
| SIZE     | W x L x H   m<br>9.225 x 9.225 x 6.6 (EH 5.0) |
| CAPA.    | m <sup>3</sup><br>561.7 (EH 425.5)            |
| HRT      | hr<br>2.0                                     |
| MAT'L    | P.C + FRP                                     |

|              |                                                                            |
|--------------|----------------------------------------------------------------------------|
| ITEM NO.     | W14-LP-4729                                                                |
| SERVICE      | LAMELLAR PLATE                                                             |
| Q'TY         | 1                                                                          |
| CAPA.        | W x L x H   m<br>BY VENDOR                                                 |
| SURFACE LOAD | 20 m <sup>3</sup> /m <sup>2</sup> .d 이하<br>(Qmax : 5,000m <sup>3</sup> /d) |
| MAT'L        | ABS+SS304                                                                  |

MAKER PACKAGE

|          |                                                  |
|----------|--------------------------------------------------|
| ITEM NO. | W14-AG-4726                                      |
| SERVICE  | BRINE 2 <sup>nd</sup><br>NEUTRALIZATION AGITATOR |
| Q'TY     | 1                                                |
| TYPE     | HYDRO PADDLE NOTE 15                             |
| SPEED    | rpm<br>(37)                                      |
| MAT'L    | STS316L/STS316L                                  |
| POWER    | kW<br>(1)                                        |

|          |                                               |
|----------|-----------------------------------------------|
| ITEM NO. | W14-AG-4727                                   |
| SERVICE  | BRINE 2 <sup>nd</sup><br>COAGULATION AGITATOR |
| Q'TY     | 1                                             |
| TYPE     | HYDRO PADDLE NOTE 15                          |
| SPEED    | rpm<br>(30)                                   |
| MAT'L    | STS316L/STS316L                               |
| POWER    | kW<br>(7.5)                                   |

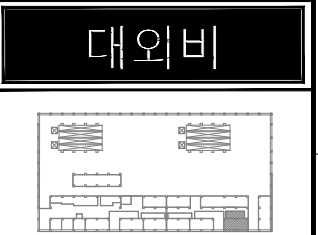
|          |                                                |
|----------|------------------------------------------------|
| ITEM NO. | W14-AG-4728                                    |
| SERVICE  | BRINE 2 <sup>nd</sup><br>FLOCCULATION AGITATOR |
| Q'TY     | 1                                              |
| TYPE     | HYDRO PADDLE NOTE 15                           |
| SPEED    | rpm<br>(25)                                    |
| MAT'L    | STS316L/STS316L                                |
| POWER    | kW<br>(3.7)                                    |

MAKER PACKAGE

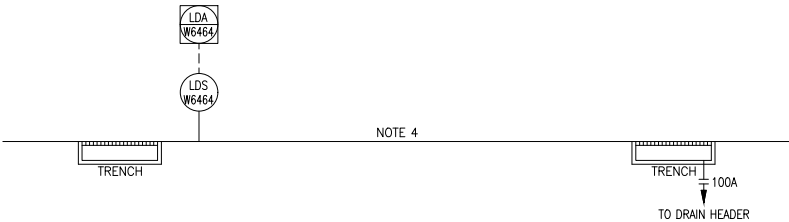
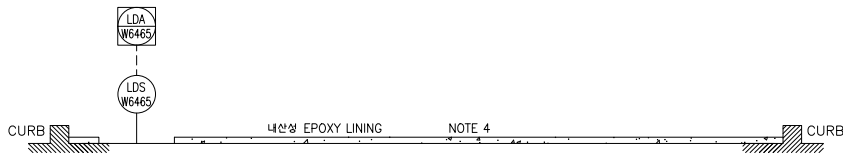
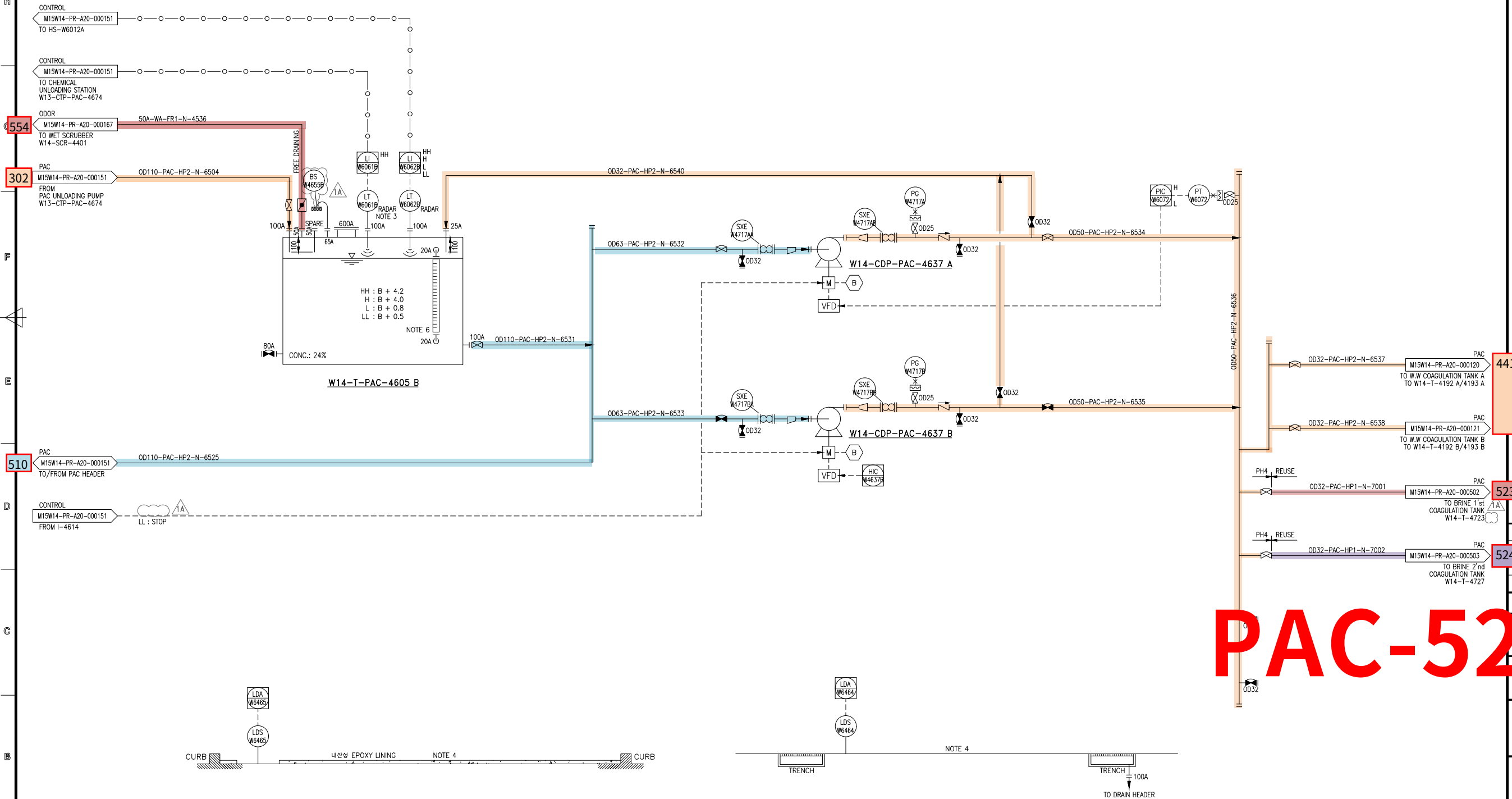
|          |                                         |
|----------|-----------------------------------------|
| ITEM NO. | W14-RD-4729                             |
| SERVICE  | BRINE 2 <sup>nd</sup><br>REDUCER DRIVER |
| Q'TY     | 1                                       |
| TYPE     | CYCLE REDUCER                           |
| CAPA.    | m <sup>3</sup> /d<br>(0.079)            |
| MAT'L    | STS304/STS304                           |
| POWER    | kW<br>(1.5)                             |

|          |                                                 |
|----------|-------------------------------------------------|
| ITEM NO. | W14-P-4729 A/B                                  |
| SERVICE  | BRINE 2 <sup>nd</sup><br>SLUDGE RETURN PUMP     |
| Q'TY     | 1 + 1                                           |
| TYPE     | CENTRIFUGAL (VVF), VORTEX                       |
| CAPA.    | m <sup>3</sup> /hr.ea x mH<br>40 x 30 (0.29MPa) |
| MAT'L    | SSC13/SSC19                                     |
| POWER    | kW<br>(30)                                      |

|          |                                |
|----------|--------------------------------|
| ITEM NO. | W14-T-PAC-4605 B               |
| SERVICE  | PAC STORAGE TANK               |
| Q'TY     | 1                              |
| TYPE     | VERTICAL & CYLINDRICAL         |
| SIZE     | ID x H m 4.0 x 4.5 (EH/4.2) 1A |
| CAPA.    | m3 57 (EH/53)                  |
| MAT'L    | FRP                            |



- NOTE
- CENTRIFUGAL PUMP  
MAT'L 표기 : CASING/IMPELLER
  - 약품종류에 따라 LINE 색상을 구분하여야 한다.
  - CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
  - CHEMICAL STORAGE TANK CURB 및 PUMP AREA에서 발생되는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.
  - 온전 시 CLOSE 되는 배관에는 LOTO 설치.
  - TANK의 역전 레벨 계지는 VENDOR에서 공급한다.



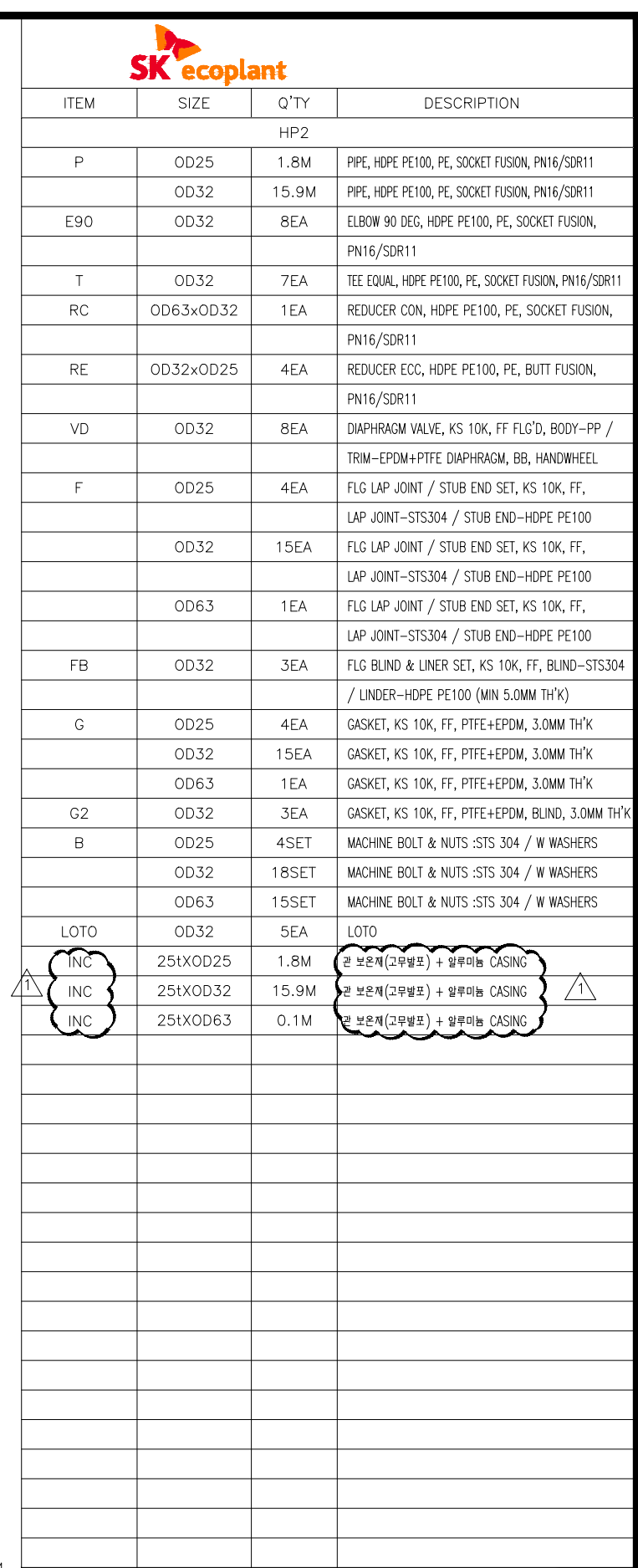
|           |                         |
|-----------|-------------------------|
| ITEM NO.  | W14-CDP-PAC-4637 A/B    |
| SERVICE   | PAC FEED PUMP           |
| Q'TY      | 1 + 1                   |
| TYPE      | CENTRIFUGAL (WVF)       |
| CAPA.     | m3/hr.ea x MPa 4 x 0.46 |
| MAT'L     | FRP/FRP                 |
| POWER  kW | 11 1A                   |

|         |                |          |
|---------|----------------|----------|
| 2A      | REVISED AFC    | 25.08.14 |
| 2       | REVISED AFC    | 25.01.17 |
| REV.    | REVISED AFC    | 24.10.31 |
| 수정 REV. | 실명 DESCRIPTION | 날짜 DATE  |

|            |     |
|------------|-----|
| 제도         | 이인학 |
| DRAWN BY   | 한진희 |
| 검토         | 박동명 |
| CHECKED BY |     |
| APPROVED   |     |

|                   |                                                                      |
|-------------------|----------------------------------------------------------------------|
| SK 에코플랜트          |                                                                      |
| 공사명 PROJECT TITLE | M15X Ph4 WWT Project                                                 |
| 도면명 DWG. TITLE    | PIPING & INSTRUMENT DIAGRAM<br>PAC STORAGE TANK & FEED PUMP<br>(2/2) |
| 축척 SCALE          | NONE                                                                 |
| 날짜 DATE           | 2025. 08. 14                                                         |
| PROJECT NO.       | 22330D                                                               |
| 도면번호 SHEET NO.    | CJ-M15W14-PR-A20-000152                                              |
| REV.              | 2A                                                                   |

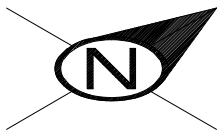




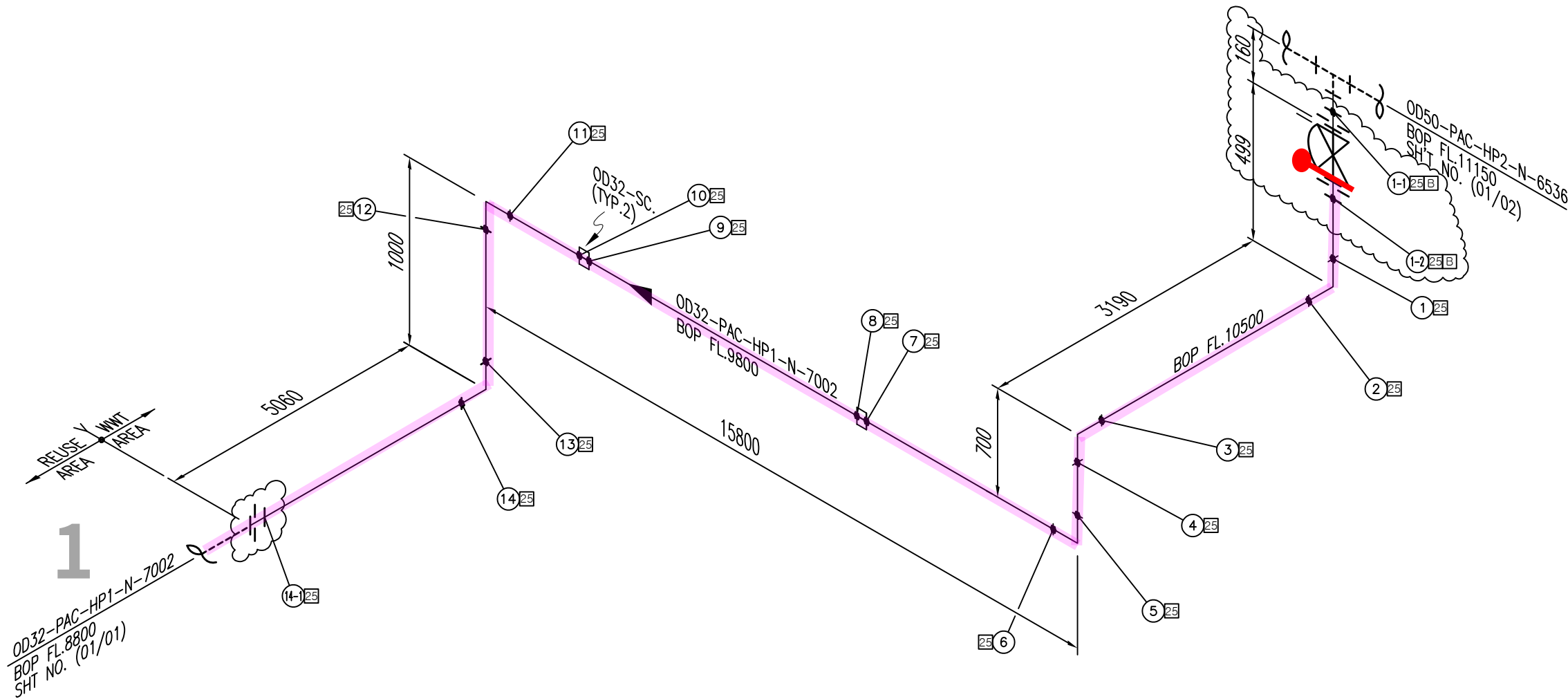
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|---------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------|
| CLIENT                                                                                |                     |  <span style="font-size: 48pt; margin-left: 20px;">2</span> |      |
| CONTRACTOR                                                                            |                     | PROJECT NAME                                                                                                                                     |      |
|  |                     | M15X Ph4 Project                                                                                                                                 |      |
| QTY                                                                                   | COMMODITY           | JOB NO.                                                                                                                                          |      |
|                                                                                       | PAC                 | 22330D                                                                                                                                           |      |
| NO.                                                                                   | LINE NO.            | SH'T NO.                                                                                                                                         | REV. |
| D                                                                                     | OD32-PAC-HP2-E-7005 | 01 / 01                                                                                                                                          | 1    |







| ITEM | SIZE | Q'TY  | DESCRIPTION                                                                        |
|------|------|-------|------------------------------------------------------------------------------------|
| HP1  |      |       |                                                                                    |
| P    | OD32 | 26.4M | PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11                                    |
| E90  | OD32 | 5EA   | ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11                            |
| SC   | OD32 | 2EA   | SOCKET, HDPE PE100, SOCKET FUSION, PN16/SDR11                                      |
| F    | OD32 | 3EA   | FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STSS304 / STUB END-HDPE PE100  |
| G    | OD32 | 2EA   | GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K                                          |
| B    | OD32 | 2SET  | MACHINE BOLT & NUTS :STS 304 / W WASHERS                                           |
| VD   | OD32 | 2EA   | DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL |



<NOTE>  
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는  
배관 ELEVATION은 현장 맞춤으로 한다.  
2. (\*) FIELD ADJUST

\* PIPE : P/P1 \* 90° ELBOW : E90 \* 45° ELBOW : E45 \* TEE : T \* RED.TEE : TR \* STUB-IN : STU \* REINFORCING PAD : RP \* CON.RED : RC \* ECC.RED : RE \* CON.SWAGE : RCS/RCS1 \* ECC.SWAGE : RES/RES1 \* NIPPLE : NP/NP1/NP2 \* CAP : CAP/CAP1 \* PLUG : PLG \* FULL COUPLING : FC \* HALF COUPLING : HF/HF1  
\* STUB-IN : STU \* SOCKET(HDPE/PVC) : SC \* SADDLE(HDPE) : SD \* FLANGE : F/F1 \* GASKET : G/G1/G2 \* BOLT & NUT : B/B1 \* BLIND FLANGE : FB/FB1 \* GATE VA : VG/VG1/VG2 \* BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO \* BUTTERFLY VA : VF/VF4 \* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7  
\* GLOBE VA : VL/VL1/VL2 \* DIAPHRAGM VA : VD/VDLO \* DAMPER : VP \* QUICK COUPLING : SQC \* FLEXIBLE JOINT(STEEL) : FLXB \* FLEXIBLE JOINT(RUBBER) : FLXR \* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL \* FLEXIBLE HOSE : FLXH \* STEAM TRAP : ST \* SPRAY NOZZLE : SPN \* EMERGENCY SHOWER : SSH  
\* STRAINER(Y-TYPE) : YSTR \* STRAINER(T-TYPE) : TSTR \* STRAINER(BUCKET-TYPE) : BSTR \* ANCHOR PLATE : STOP \* BIRD SCREEN : BS \* ISOLATION KIT : IK \* UNLOADING BOX : UB \* AIR TRAP : AT \* LOTO : LOTO \* SIGHT GLASS : SG \* PRESSURE GAUGE : PG \* SHILED : SH \* SPECIAL BOLT & NUT : BL/BL1

CLIENT

3

|            |        |   |           |                     |               |                                        |     |                                                                                       |          |                           |     |     |       |                                                                                       |                     |          |      |
|------------|--------|---|-----------|---------------------|---------------|----------------------------------------|-----|---------------------------------------------------------------------------------------|----------|---------------------------|-----|-----|-------|---------------------------------------------------------------------------------------|---------------------|----------|------|
| INSULATION | CODE   | N | CONDITION | PRESS.<br>(kg/cm2g) | TEMP.<br>(°C) | AREA                                   | W14 |                                                                                       |          |                           |     |     |       | CONTRACTOR                                                                            | PROJECT NAME        |          |      |
|            | THICK. | — | DESIGN    | 1                   | 60            | UNIT                                   | 2FA |                                                                                       |          |                           |     |     |       |  | M15X Ph4 Project    |          |      |
|            |        |   | OPERATING | ATM                 | AMB           | REF. PLAN DWG.                         |     |                                                                                       |          |                           |     |     |       | COMMODITY                                                                             | JOB NO.             |          |      |
|            |        |   |           |                     |               | CJ-M15W14-PI-A30-020011-H              |     |  | 24.12.20 | APPROVED FOR CONSTRUCTION | 황래중 | 안남후 | 박동영   | LINE NO.                                                                              | OD32-PAC-HP1-N-7002 | SH'T NO. | REV. |
|            |        |   |           |                     |               | REF. P & ID<br>CJ-M15W14-PR-A20-000152 |     | REV.<br>NO.                                                                           | DATE     | DESCRIPTION               | BY  | CHK | APP'D |                                                                                       |                     | 01 / 01  | 0    |